BIONTECH

Company analysis and valuation

AC444 – Summative Assessment by Group 1, Seminar 1

London, 31st of March 2025

Team

Experienced and passionate team with proven capabilities

BIONTECH

Raphael Benesch

Luca Beutin

Shiv Handa

Kilian Kremp

Maximillian Leitz

Tanveer Sira

Advising corporate clients

Auditing, transaction services, and valuation

Growth and buy-out investing

BARCLAYS

BCG

Bridgepoint

EY

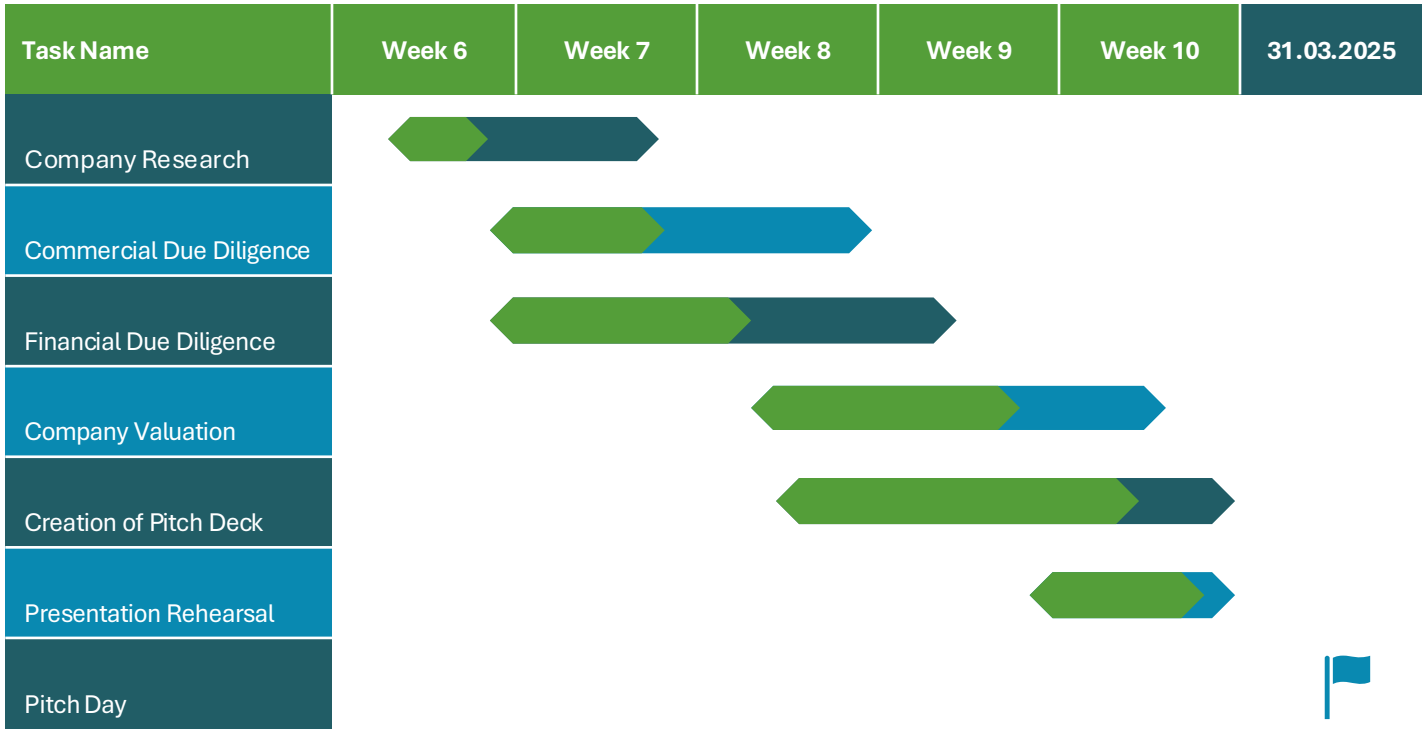
J.P.Morgan

pwc

Project overview and key challenges

Continuous progress along the project timeline ensured early engagement to solve challenging situations

Project timeline



- Tasks like **Company Research, Due Diligence, and Valuation are scheduled sequentially** to build upon one another.
- **Foundational insights from early research feed into due diligence efforts**, which in turn inform valuation.
- Commercial and Financial Due Diligence are run in parallel during Weeks 6–8 **to save time without compromising quality.**

Key challenges

Negative cash flows prior & post COVID-19 paired with heavy R&D spending

a. Prior to 2020, BioNTech's financial statements had **very limited data depth** compared to its larger peers
b. This created **explicit challenges when standardising financial statements** to make different companies comparable
c. When forecasting cash flows, **heavy R&D spending, funded by large cash balances**, kept the company burning through cash

Finding accounting adjustments that are not COVID-19 related

a. Upon **comparing the accounting policies** of BioNTech and all chosen peers, it became apparent that there were no viable accounting adjustments
b. The **limited data, especially on research efforts and capitalisation**, further limited the availability of viable adjustments

Forecasting BioNTech's performance

a. Given BioNTech's **singular product portfolio**, future performance heavily depends on R&D success and innovation
b. Creating an **extensive revenue model based on current product trials** across different phases was necessary to make reasonable assumptions concerning company valuation
c. **Forecasting R&D spending, a key future revenue driver**, therefore exceeded revenue in the near term

Agenda for today

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- Business overview and strategy
- ESG performance analysis
- Financial projections
- Ratio analysis
- Accounting adjustments
- Valuation

Business overview and strategy

BioNTech at a glance: company overview

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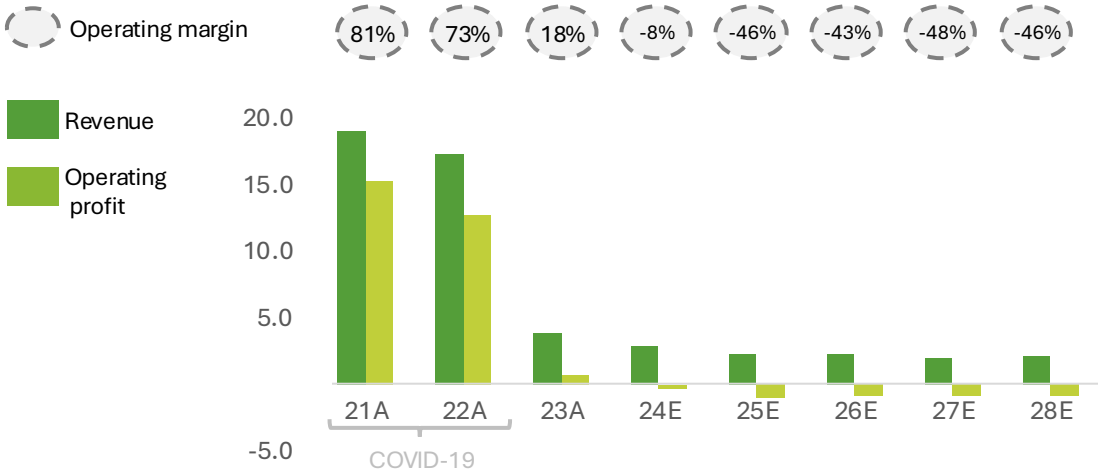
German biotechnology company specialising in the development of mRNA-based immunotherapies

Company information

- BioNTech employs **6,300+ people** and is **headquartered in Mainz, Germany**
- It operates in the rapidly growing **biotechnology and immunotherapy market**, **focusing on mRNA-based vaccines, oncology treatments, infectious diseases, and personalised medicine**
- **Founded in 2008**, BioNTech gained global recognition for developing the **first mRNA COVID-19 vaccine** in collaboration with Pfizer
- The company is **publicly traded (Nasdaq: BNTX),** with key investors including founders, institutional investors, and strategic partners like Pfizer
- BioNTech has a **strong international presence**, with **multiple R&D centres, advanced manufacturing facilities, and a growing pipeline of innovative treatments**
- **Increased global demand for mRNA technology and personalised medicine** is driving BioNTech's expansion into **cancer immunotherapy, infectious diseases, and autoimmune treatments**
- The company integrates **sustainability and cutting-edge biopharmaceutical innovations**, aiming to expand mRNA applications while ensuring efficient, scalable production and global accessibility

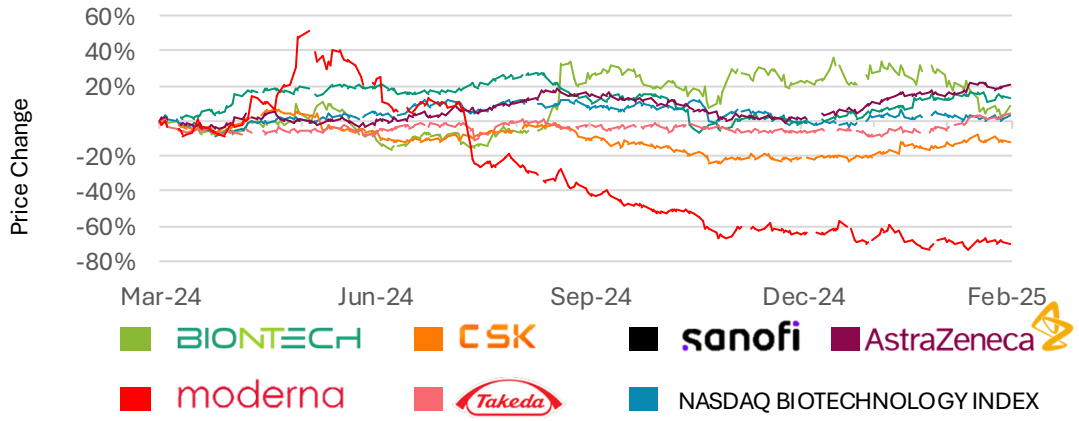
Key financials 21A-28E (EURbn)



Partnerships

Partnerships	Sustainable impact (UN SDG)
Autolus, Genentech, CEPI, Bill & Melinda Gates foundation, Pfizer	3 Good health and well-being; 9 Industry, innovation and infrastructure; 17 Partnerships for the goals

Stock performance vs. market index (in %, Mar-24 to Feb-25)



Key management

Co-Founder & CEO – Ugur Sahin

Next to his role at BioNTech is a Prof. for translational oncology and immunology

CFO – Jens Holstein

Prior to joining BioNTech, Jens was CFO of MorphoSys AG and is a trained consultant

COO – Annemarie Hanekamp

Previously, held various leadership positions at Novartis and Bristol Myers Squibb

Chair of Board – Helmut Jeggle

Next to his role at BioNTech, he holds several board positions within the biotech space

Source(s): Capital IQ as of 25th of March 2025, BioNTech Annual Report 2023.

Product overview

Active immunotherapies for patient-specific approaches to treating serious diseases

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Oncology

- **mRNA-based therapies:** Personalised cancer immunotherapies that use mRNA to instruct patient cells to produce tumour-specific antigens, triggering a targeted immune response against cancer cells
- **Cell therapies:** Engineering patient-derived T-cells to recognise and attack cancer cells more effectively
- **Antibody-based therapies:** Developing monospecific and bispecific antibodies, as well as antibody-drug conjugates (ADCs), to target various tumour types through distinct mechanisms

Infection Diseases

- **COVID-19:** Developed Comirnaty (tozinameran) in collaboration with Pfizer, the first approved mRNA-based vaccine, widely used during the COVID-19 pandemic
- **Influenza:** Advancing mRNA-based influenza vaccines, though a recent combination COVID-19 and influenza vaccine did not fully meet late-stage trial goals, particularly against influenza B
- **Malaria and tuberculosis:** Conducting research to develop mRNA vaccines for these globally significant infectious diseases, aiming to enhance prevention efforts

Autoimmune Diseases

- **mRNA-based immune tolerance:** Developing non-inflammatory mRNA vaccines to retrain the immune system and prevent attacks on self-antigens
- **Pipeline expansion:** Extending mRNA expertise beyond oncology and infectious diseases into chronic immune-mediated disorders
- **Multiple Sclerosis (MS)**: Preclinical mRNA therapy showed restored motor function and reduced inflammation in EAE (mouse model of MS) by expanding regulatory T cells

Source(s): BioNTech Annual Report 2023

Competitive forces shaping the biotech industry

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Innovation-driven market environment fuels intensive competition among market participants

Competitors

- **Strong competition** from firms such as Moderna, Pfizer or AstraZeneca
- **MRNA vaccine space is expanding**, with multiple companies investing in similar technologies
- Large pharma firms with established **R&D, distribution, and production capabilities** pose a **significant challenge**
- Continuous **innovation is required** to stay competitive
- Market differentiation relies on **R&D advancements, partnerships, & IP protection**

HIGH

New entrants

- **High barriers to entry** due to significant R&D costs, regulatory approvals, and patent protection
- Long drug development cycles and need for clinical trials make it **difficult for new players to enter quickly**
- **Strong industrial regulations** (e.g. FDA and EMA) make compliance challenging for new entrants
- However, advancements in biotech, AI/driven drug discovery, and VC investments could enable **new startups to enter specific niche areas**

MODERATE

Suppliers

- Key suppliers include **raw material providers for mRNA production, lipids, and bioreactors**
- Dependence on a **limited number of specialised suppliers** for lipid nanoparticles and cold-chain logistics raises risk
- BioNTech has **strong negotiation power** due to scale and established partnerships
- Supply chain disruptions, such as Covid/19 pandemic, highlight **vulnerabilities in securing critical components**

LOW - MODERATE

Substitutes

- **Traditional vaccine platforms** (e.g., protein-based or viral vector vaccines) can serve as substitutes
- **Alternative cancer treatments** (e.g., radiation therapy) may replace BioNTech's immunotherapies
- **Rapid development in gene editing and personalised medicine** could shift treatment approaches
- New mRNA competitors may develop **faster, cheaper, or more effective solutions**, impacting BioNTech's market share

HIGH

Customers

- **Governments and healthcare organisations** have **moderate to high bargaining power** due to bulk purchasing and alternative vaccine suppliers
- **Individual patients have low bargaining power** as they rely on approved and available treatments
- For life-saving therapies, **patients follow doctor prescriptions**, limiting their choice
- National healthcare systems often cover costs, **reducing individual influence**

LOW - MODERATE

Note: Assessment of competitive threat

BioNTech's market positioning and differentiation

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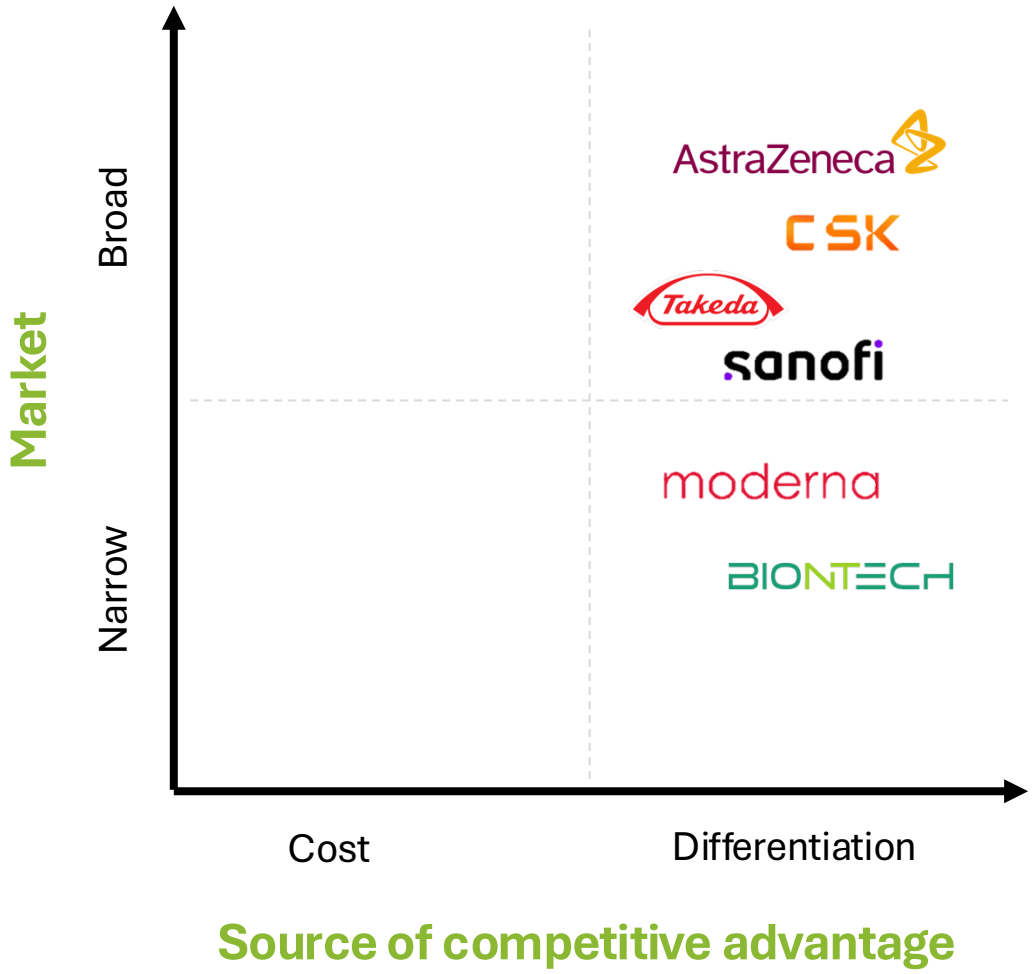
Focus on innovation by targeting specialised therapies to pivot away from covid vaccines



Differentiation

High R&D investment: BioNTech reinvests a significant portion of revenue into next-gen mRNA technology, AI-driven drug discovery, and clinical trials, prioritising innovation over cost efficiency

Narrow markets

Narrow market: Personalised mRNA cancer immunotherapies focus on individualised treatments, requiring biomarker-driven customisation and serving specialised patient groups

Overall market positioning: narrow – Long-term strategy leans toward specialised, high-value therapies in oncology and autoimmune diseases, making it primarily a narrow-market company

Core competencies

First-mover advantage & vertical integration: BioNTech leads in mRNA innovation, securing a strong IP portfolio, while its vertically integrated manufacturing enhances control over R&D, production, and distribution

Risks & challenges

Revenue dependence & regulatory hurdles: BioNTech faces overreliance on COVID-19 revenues, clinical and regulatory challenges in oncology, and IP litigation risks that could impact long-term growth

ESG performance analysis

Overview of BioNTech's ESG performance

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Innovation driven market environment fuels intensive competition among market participants

ESG factors	Sustainability initiatives and achievements	Impact on operations	Risks and considerations
Environm. & climate protection (E)	**Target of CO^2 10% reduction** by 2024 **Investment in sustainable technologies** for resource efficiency (water & electricity) **Commitment to sustainable supply chain** and **waste reduction**	1. Sustainable production processes via renewables adoption, &energy-efficient tech. 2. Development of responsible sourcing practices 3. Investment in waste reduction, water conservation, and low-carbon man. facilities	▪ **Manufacturing and supply chain volatility** due to increased exposure to climate change risks ▪ **Investment in regulatory compliance** due to increasing environmental regulations ▪ **Strong margin impact of costs** due to dependencies on sustain. raw materials & energy
Attractive employer (S)	**Fostering an inclusive workforce** through employee equity programmes and DE&I initiatives **Community development** by investing in global health (incl. underserved regions) **Workplace welfare** through mental health and work-life balance initiatives	1. Strengthened workplace culture, improved employee retention and satisfaction 2. Greater social impact 3. Increase in workforce engagement	▪ **Continuous workforce upskilling** to align with technological advancements ▪ **Ineffective DE&I program** leading to equity & inclusion gaps across different regions. ▪ **High employee well-being expectations** requiring additional investments
Responsible governance (G)	**High ESG ratings** achieved (prime status from ISS ESG ~top 10%) **Independent board oversight** and **accountability mechanisms** **Ethical business practices**	1. Reinforced corporate reputation and investor confidence 2. Ensured transparency and accountability 3. Enhanced stakeholder engagement and responsible corporate reporting	▪ **Investment in regulatory compliance** due to increased regulatory scrutiny ▪ **Enhanced cybersecurity & data protection** measures required through external threats ▪ **Continuous reinforcement of ethical standards** across global operations

Source(s): BioNTech Sustainability Report 2023

Performance-driving ESG factors (1/2)

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BioNTech is on good track of achieving the relevant environmental & social performance drivers

Driver	Climate change & decarbonisation commitments	Ethical medicine access & global health equity
Description	BioNTech has committed to reducing its carbon footprint through validated Science-Based Targets (SBTi). The firm integrates renewable energy, low-emission manufacturing processes, and sustainable packaging.	BioNTech is expanding affordable & innovative vaccines to low- and middle-income countries. The firm has developed the BioNTainer system, a modular vaccine manufacturing facility for mRNA vaccine production in Africa.
Accounting analysis	a. **Capitalisation** of carbon reduction investments b. **Increased operating expenses** for energy-efficient manufacturing c. **Asset impairment** for high emission production sites	a. **Provisions** for low-cost distribution in emerging markets b. **Increased R&D spending** on equitable healthcare solutions c. **Creation of liabilities** through global health partnerships
Financial & ratio analysis	a. **Higher CapEx** for renewables investments b. **Improved profitability margin** (long-term) due to energy efficiency c. **Impact on solvency ratios** due to pot. financing arrangement	a. **Lower profit margins** due to equitable pricing strategies in emerg. markets b. **Improved revenue stability** from LT public health contracts c. **Increase in CFO** due to funding from global health agencies
Strategy analysis	▪ Moat by meeting investor/consumer demand for sustainable biotech ▪ Risk mitigation against taxes, SC disruption, and regulations ▪ R&D focus on sustainable pharma processes and clean manufacturing	▪ Expansion into emerging markets as LT strategy mitigating reliance on West ▪ Moat as ESG-first biotech company prioritising health equality ▪ Strong brand reputation attracting sustainable investors and consumers
Prospective analysis & valuation	1. **Lower discount rate** (WACC) due to preferred green financing 2. **Higher valuation multiples** due to low-carbon stock preference of green investors	1. **Higher valuation multiples** for sustainable biotech firms 2. **Risk premium reduction** due to revenue diversification & state-backing
Current progress	65%	75%

Performance-driving ESG factors (2/2)

Key governance drivers are multi-faceted and at different development stages

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Driver	Data privacy & bioethics in AI-driven biotech	Regulatory compliance & ESG reporting standards
Description	BioNTech is integrating AI-driven analytics to enhance drug discovery and personalised medicine, ensuring compliance with data security regulations and bioethics governance.	BioNTech is strengthening governance through transparent ESG reporting and compliance with evolving regulatory frameworks.
Accounting analysis	a. **Compliance costs** for data security regulations (GDPR) b. **Capitalisation** of AI-based analytics R&D investments (InstaDeep) c. **Legal liabilities** related to bioethics and AI governance	a. **Adjustments** in accounting frameworks for double materiality b. **Increased internal auditing costs** for ESG disclosures c. **Compliance costs** for EU CSRD and ESRS regulations
Financial & ratio analysis	a. **Increased R&D expenses** for secure AI applications in biotech b. **Lower legal risk provisions** due to strong corporate governance c. **Strong liquidity ratios** from AI-driven efficiency in drug discovery	a. **Short-term compliance costs** impact net income margins b. **Long-term cost reductions** via standardised ESG reporting c. **Improved credit ratings** from transparent ESG disclosures
Strategy analysis	▪ AI integration in cancer immunotherapy reduces drug development time ▪ Governance frameworks ensure ethical AI use in patient centric treatment ▪ Competitive advantage through secure, AI powered biotech innovation	▪ Proactive governance framework to meet EU sustainability regulations ▪ ESG leadership attracts compliance-focused institutional investors ▪ Enhanced supply chain due diligence mitigates non-compliance risks
Prospective analysis & valuation	1. **Increased revenue projections** and long-term cash flow stability 2. **Reduced cost of capital and higher valuation multiples** due to lower regulatory risk	1. **Premium valuations** vs biotech peers due to strong ESG compliance 2. **Reduced volatility in earnings forecasts** due to lower risk exposure
Current progress	50%	70%

Financial projections

Condensed income statement

BioNTech

BioNTech's revenues surged in 2021 and 2022 as a result of its COVID-19 vaccine success

	2017A	2018A	2019A	2020A	2021A	2022A	2023A
INCOME STATEMENT							
1 **Revenue**	**61.6**	**127.6**	**108.6**	**482.3**	**18,976.7**	**17,310.6**	**3,819.0**
Operating expenses	(122.9)	(181.4)	(290.1)	(564.7)	(3,692.9)	(4,667.9)	(3,128.6)
Operating profit	**(61.3)**	**(53.9)**	**(181.5)**	**(82.4)**	**15,283.8**	**12,642.7**	**690.4**
Investment Income	-	-	-	-	-	216.8	162.0
Interest income	2.1	8.0	4.1	1.6	67.7	113.5	357.6
Interest expense	(26.8)	(1.9)	(2.0)	(65.0)	(305.1)	(18.9)	(23.9)
Profit before taxes	**(85.9)**	**(47.7)**	**(179.4)**	**(145.8)**	**15,046.4**	**12,954.1**	**1,186.1**
Tax expense	(0.0)	(0.6)	0.2	161.0	(4,753.9)	(3,519.7)	(255.8)
Profit after taxes	**(85.9)**	**(48.3)**	**(179.2)**	**15.2**	**10,292.5**	**9,434.4**	**930.3**
Profit/loss to non-controlling interest	**0.3**	**0.2**	**0.0**	**-**	**-**	**-**	**-**
Profit/Loss of discontinued operations	-	-	-	-	-	-	-
2 **Profit/loss to ordinary shareholders**	**(85.7)**	**(48.0)**	**(179.2)**	**15.2**	**10,292.5**	**9,434.4**	**930.3**
3 **OPERATING EXPENSES**							
Classification by function:							
Cost of sales (by function)	(9.3)	(13.7)	(17.4)	(59.3)	(2,911.5)	(2,995.0)	(599.8)
SG&A (by function)	(30.1)	(29.4)	(48.2)	(108.5)	(327.2)	(541.2)	(557.7)
R&D expenses	(85.5)	(143.0)	(226.5)	(645.0)	(949.2)	(1,537.0)	(1,783.1)
Other operating income (by function)	2.1	4.7	2.0	248.1	495.0	405.3	(188.0)
Classification by nature:							
Depreciation and amortization (by nature)	(10.5)	(22.0)	(33.9)	(38.7)	(75.2)	(123.3)	(183.4)

1

- **Revenues** surged from €482.3m in 2020 to €19.0bn and €17.3bn in 2021 and 2022. This was driven by BioNTech's success with the COVID-19 vaccines
- With comparatively low **operating expenses** of 19% and 27% (as % of revenues), BioNTech achieved **operating profits** of €15.3bn and €12.6bn in 2021 and 2022

2

- During the Covid-19 peak, BioNTech achieved **net profits** of €10.3bn and €9.4bn. In 2023, net income declined to €930.3m as a result of waning demand for Covid-19 vaccines
- **Finance income** of €330.3m and €519.6m (pre-tax) from the company's large cash balances also contributed to BioNTech's net profits in 2022 and 2023

3

- The company's **operating expenses** comprise of cost of sales, SG&A and **R&D expenses,** with the latter contributing the most with €1.8bn in 2023, up from €1.5bn in 2022
- Classified by nature, BioNTech also recognised increasing charges for the **depreciation and amortisation** of its assets, totalling €183.4m in 2023

Condensed balance sheet

BioNTech increased its asset base from €1.7bn in 2020 to €19.3bn in 2023

	2017A	2018A	2019A	2020A	2021A	2022A	2023A
BALANCE SHEET							
ASSETS							
Non-current tangible assets	101.5	116.0	148.1	327.0	534.8	827.6	1,055.0
Non-current intangible assets	83.5	88.0	89.4	163.5	202.4	219.7	1,166.6
Other non-operating investments	0.0	0.0	-	-	21.3	80.2	1,176.1
Deferred tax asset	0.7	0.9	0.8	162.1	0.4	230.0	260.4
Total non-current assets	**185.8**	**204.9**	**238.3**	**652.6**	**758.9**	**1,357.5**	**3,658.1**
Inventories	3.9	5.8	11.7	64.1	502.5	439.6	357.7
Trade receivables	4.6	18.9	11.9	165.5	12,381.7	7,145.6	2,155.7
Other current assets	8.4	11.8	16.6	226.2	495.0	461.3	5,171.1
Cash and cash equivalents	172.1	411.5	519.1	1,210.2	1,692.7	13,875.1	11,663.7
Total current assets	**188.9**	**448.1**	**559.4**	**1,666.0**	**15,071.9**	**21,921.6**	**19,348.2**
Total assets	**374.7**	**653.0**	**797.6**	**2,318.6**	**15,830.8**	**23,279.1**	**23,006.3**
LIABILITIES AND SHAREHOLDERS' EQUITY							
Ordinary shareholders' equity	**(49.3)**	**266.2**	**493.5**	**1,371.8**	**11,893.7**	**20,055.6**	**20,245.9**
Non-controlling interest in equity	1.1	0.8	-	-	-	-	-
Non-current debt	-	-	68.9	231.0	171.6	176.2	191.0
Deferred tax liability	-	-	-	0.2	66.7	6.2	39.7
Other non-current liabilities (non interest bearing)		264.4	259.9	97.1	109.6	212.8	80.1
Non-current liabilities	**264.4**	**259.9**	**166.0**	**340.8**	**451.1**	**262.5**	**689.9**
Trade payables	52.5	41.7	20.5	102.3	160.0	204.1	354.0
Other current liabilities	106.0	84.4	112.3	494.6	3,196.1	2,720.9	1,688.4
Current debt	-	-	5.3	9.1	129.9	36.0	28.1
Current liabilities	**158.5**	**126.1**	**138.1**	**606.0**	**3,486.0**	**2,961.0**	**2,070.5**
Total liabilities and shareholders' equity	**374.7**	**653.0**	**797.6**	**2,318.6**	**15,830.8**	**23,279.1**	**23,006.3**

1

- BioNTech's **tangible assets** comprise of **PP&E** of €757m and **right of use assets** of €241m in 2023
- The intangible assets include primarily concessions, licenses, in-process R&D, and **goodwill** of €782m and €363m as of 2023
- Further, **deferred tax assets** of €260.4m and **non-operating investment assets** of €1.2bn were recorded in 2023

2

- At the same time as BioNTech's strong increase in sales, its **net working capital** increased similarly, peaking at €12.4bn in 2021 before declining again in the following two years
- The company's successful years 2021 and 2022 are also reflected in a large **cash and cash equivalents** balance, amounting to €11.7bn in 2023, mostly invested in money market funds

3

- In relation to its total assets, BioNTech only has a small portion of **current and non-current debt**, with its interest-bearing liabilities amounting to 291m in 2023
- The other current and non-current liabilities of over €2.1bn consist primarily of contract liabilities, income tax liabilities and financial liabilities

Oncology pipeline offsets COVID-19 declines

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COVID-19 vaccine sales decline while oncology (BNT122 & BNT116) emerges as new revenue stream

Revenue by type (EUR m)	2024E	2025E	2026E	2027E	2028E	2029E
1 **Existing product sales** (Comirnaty COVID-19)	**2,750**	**2,200**	**1,870**	**1,590**	**1,351**	**1,351**
Volume decline		*-20%*	*-15%*	*-15%*	*-15%*	
Pipeline product sales – oncology			**113**	**225**	**416**	**563**
BNT122 TAM (cases m)			0.15	0.15	0.15	0.15
Penetration (share)			*5.0%*	*10.0%*	*17.0%*	*22.0%*
BNT122 SOM (cases m)			0.0075	0.015	0.0255	0.033
BNT122 pricing			50000	50000	50000	50000
BNT122 PoS			*30%*	*30%*	*30%*	*30%*
2 BNT122 sales			112.5	225	382.5	495
BNT116 TAM (cases m)					0.1	0.1
Penetration (share)					*3.0%*	*6.0%*
BNT116 SOM (cases m)					0.003	0.006
BNT116 pricing					75000	75000
BNT116 PoS					*15%*	*15%*
3 BNT116 sales					33.8	67.5

1

- **Comirnaty (COVID-19) sales decline** from €2,750m in 2024E to €1,351m by 2029E
- Reflects **15–20% annual decline** as pandemic demand fades
- Establishes a lower, more sustainable revenue base
- Underscores the need for **pipeline assets to offset lost volume**

2

- **BNT122 sales increase** from €112.5m in 2026E to €495m by 2029E
- Driven by **penetration growth from 5.0% to 22.0%** of a stable **150k TAM**
- Pricing remains at **€50k**, with **PoS of 30%**, reflecting conservative assumptions
- Early revenue ramp supports BioNTech's **transition into commercial oncology**

3

- **BNT116 pot. adds** €33.8m in 2028E and €67.5m in 2029E
- Entering late in the forecast due to **early stage of FDA approval**
- Represents an estimated **TAM** of **100k cases p.y.** with growth potential
- Expands the oncology portfolio beyond a single asset and **diversifies risk**

Note: TAM = Total Addressable Market, SOM = Serviceable Obtainable Market, PoS = Probability of Success

New vaccines expand revenue mix

Flu vaccines are likely to diversify sales, while success of shingles vaccines still uncertain

Revenue by type (EUR m)	2024E	2025E	2026E	2027E	2028E	2029E
Pipeline product sales – new vaccines			**45**	**127**	**239**	**292**
Shingles vaccine market					5,300	6,000
Pricing					150	150
Shingles TAM (doses m)					35	40
Penetration (share)					*5%*	*5%*
Shingles SOM (doses m)					1.8	2
Shingles pricing					150	150
Shingles PoS					*25%*	*25%*
Shingles sales					66	75
Flu TAM (doses m)			150	150	150	150
Penetration (share)			*2.0%*	*5.7%*	*7.7%*	9.7%
Flu SOM (doses m)			3	8.49	11.49	14.49
Flu pricing			30	30	30	30
Flu PoS			*50%*	*50%*	*50%*	*50%*
Flu sales			45	127	172	217
Partnership & milestone revenue	**50**	**50**	**150**	**50**	**50**	**50**
Collaboration revenue	50	50	50	50	50	50
Milestone bump			100			
Total revenue	**2,800**	**2,250**	**2,178**	**1,992**	**2,056**	**2,256**

Note: TAM = Total Addressable Market, SOM = Serviceable Obtainable Market, PoS = Probability of Success

1

- **Shingles sales reach €66m in 2028, rising to €75m in 2029E**
- Reflects **5% market penetration** of a **2m dose SOM**, adjusted for **25% PoS**
- Backed by consistent **€150 pricing**, targeting a **large adult immunisation market**
- Provides **moderate but reliable** contribution to portfolio revenue

2

- **Flu vaccine revenue increases** from €45m in 2026E to €217m by 2029E
- Driven by **market penetration** to 11–14 million doses by 2029
- **Stable €30 per dose pricing** and **50% PoS** support probable scale-up
- Positions BioNTech in the **large and recurring vaccine segment**

3

- Revenue declines from **€2,800m in 2024** to **€1,992m in 2027**, as COVID-19 sales taper off
- **€50m annual collaboration** fees provide steady baseline income (incl. €100m milestone)
- Pipeline expansion lifts revenue to **€2,256m by 2029**, reflecting a shift to diversified growth

Cost structure stabilises and efficiency improves

SG&A and COGS remain steady, investments in R&D, and tighter cash conversion cycle

BIONTECH

	2019A	2020A	2021A	2022A	2023A	2024E	2025E	2026E	2027E	2028E
Assumptions										
Income Statement										
Revenue growth (% YoY)	-14.9%	344.1%	3834.6%	-8.8%	-77.9%	-26.7%	-19.6%	-3.2%	-8.5%	3.2%
Cost of goods sold (% of revenue)	-16.0%	-12.3%	-15.3%	-17.3%	-15.7%	-15.3%	-15.3%	-15.3%	-15.3%	-15.3%
SG&A (% of revenue)	-44.4%	-22.5%	-1.7%	-3.1%	-14.6%	-14.6%	-14.6%	-14.6%	-14.6%	-14.6%
R&D (% of revenue)	208.6%	133.7%	5.0%	8.9%	46.7%	80.0%	117.6%	114.8%	120.2%	117.7%
Other operating income / expenses (% of revenue)	1.8%	51.4%	2.6%	2.3%	-4.9%	2.0%	2.0%	2.0%	2.0%	2.0%
Other										
Tax rate						30.0%	30.0%	30.0%	30.0%	30.0%
Working Capital Schedule										
Accounts receivable (days)	51	66	119	203	438	59	59	59	59	59
Inventory (days)	181	230	35	57	239	46	46	46	46	46
Accounts payable (days)	644	373	16	22	167	19	19	19	19	19
Cash conversion cycle	**(411)**	**(76)**	**138**	**238**	**510**	**85**	**85**	**85**	**85**	**85**
Operating working capital										
Trade receivables	12	166	12,382	7,146	2,156	450	362	350	320	330
Inventories	12	64	503	440	358	54	43	42	38	40
Trade payables	42	20	102	160	204	22	18	17	16	16
= Ending operating working capital	**(18)**	**209**	**12,782**	**7,425**	**2,309**	**482**	**387**	**375**	**343**	**354**
Change in NWC (cash inflow / (outflow))		*(227)*	*(12,573)*	*5,357*	*5,116*	*1,828*	*95*	*12*	*32*	*(11)*

1
- **Revenues fell by 77.9% in 2023**, reflecting a sharp decline from the post-pandemic peak
- **Stabilisation begins in 2024**, with revenue decline moderating
- **Growth returns by 2028**, with a projected **+3.2% YoY increase**

2
- **COGS** remains **constant** at **15% of revenue**
- **SG&A has normalised**, dropping from higher levels in early stage to around 14–15%

3
- **R&D** spending forecast **exceeds revenue**
- Highlights BioNTech's commitment to **reinvesting pandemic windfalls** into long-term **pipeline development**

4
- DSO drops and Inventory Days fall, **shrinking** the **Cash Conversion Cycle** from 138 to 85 days

5
- **Lower working capital needs** are expected to continue to **free up liquidity** through to 2028

Ratio analysis

BioNTech and industry peers' ratio analysis

Overview of financial ratios for pre and after COVID-19

	BioNTech		Astra Zeneca		Moderna		Sanofi		GSK		Takeda	
Ratio analysis	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**
Decomposition of ROE												
Net profit margin (ROS)	3%	24%	12%	13%	-93%	-69%	33%	12%	21%	18%	6%	4%
× Asset turnover	0.31	0.17	0.43	0.47	0.19	0.31	0.33	0.32	0.30	0.51	0.26	0.29
= Return on assets (ROA)	1%	4%	5%	6%	-18%	-21%	11%	4%	6%	9%	1%	1%
× Financial Leverage	1.67	1.15	4.24	2.59	2.36	1.34	1.85	1.69	4.10	5.20	2.66	2.15
= Return on equity (ROE)	**2%**	**5%**	**21%**	**16%**	**-42%**	**-29%**	**20%**	**7%**	**26%**	**47%**	**4%**	**3%**
Decomposition of RNOA												
Net Operating Profit Margin	7%	20%	12%	13%	-93%	-69%	32%	12%	21%	18%	6%	4%
× Net Operating Asset Turnover	2.01	0.51	0.83	0.71	-0.73	2.67	0.49	0.48	0.54	1.03	0.34	0.36
= Return on Net Operating Assets	**15%**	**10%**	**10%**	**9%**	**68%**	**-184%**	**16%**	**6%**	**11%**	**18%**	**2%**	**2%**
Other key profitability ratios												
Gross profit margin	*88%*	*84%*	*80%*	*82%*	*-72%*	*-39%*	*68%*	*69%*	*70%*	*72%*	*64%*	*68%*
EBITDA margin	*-25%*	*27%*	*7%*	*7%*	*-119%*	*-60%*	*31%*	*8%*	*16%*	*15%*	*6%*	*6%*
Asset management ratios												
PP&E turnover	2.03	4.06	2.90	4.21	2.46	2.87	3.31	3.31	1.96	2.74	1.99	1.96
Days' receivables	66.21	438.40	48.62	60.98	291.83	60.72	74.31	73.81	83.40	66.97	89.05	62.95
Days' inventories	230.15	239.27	239.99	225.22	5.72	26.03	243.04	265.69	299.60	226.51	236.30	286.05
Days' payables	372.74	167.49	136.99	129.31	3.13	18.90	157.73	201.71	213.13	161.28	96.01	117.95
Cash conversion cycle	**-76.38**	**510.18**	**151.62**	**156.90**	**294.42**	**67.86**	**159.63**	**137.79**	**169.87**	**132.20**	**229.34**	**231.05**
Solvency ratios												
Debt-to-equity	0.17	0.01	1.62	0.83	0.10	0.07	0.46	0.29	1.66	1.91	1.18	0.79
Debt-to-invested capital	0.14	0.01	0.62	0.45	0.09	0.07	0.32	0.23	0.62	0.66	0.54	0.44
Interest coverage (earnings based)	0.24	-51.63	-7.31	-7.66	72.40	101.74	-32.80	-12.38	-8.13	-10.54	-2.55	-2.89
Sustainable growth rate												
ROE	2%	5%	21%	16%	-42%	-29%	20%	7%	26%	47%	4%	3%
× Dividend payout ratio	0%	0%	114%	75%	0%	0%	32%	87%	0%	0%	157%	162%
Sustainable growth rate	**2%**	**5%**	**-3%**	**4%**	**-42%**	**-29%**	**14%**	**1%**	**26%**	**47%**	**-2%**	**-2%**

BioNTech ratio analysis: key highlights (1/2)

Strong performance of BioNTech during pandemic reverts to industry averages

Ratio analysis	Industry avg.* 2019A	BioNTech 2019A	Industry avg.* 2020A	BioNTech 2020A	Industry avg.* 2021A	BioNTech 2021A	Industry avg.* 2022A	BioNTech 2022A	Industry avg.* 2023A	BioNTech 2023A
Decomposition of ROE										
Net profit margin (ROS)	7%	-165%	18%	3%	11%	54%	13%	55%	12%	24%
× Asset turnover	0.39	0.15	0.33	0.31	0.34	2.09	0.38	0.89	0.40	0.17
= Return on assets (ROA)	3%	-25%	6%	1%	3%	113%	5%	48%	5%	4%
× Financial leverage	3.76	1.91	3.21	1.67	2.79	1.37	2.76	1.22	2.91	1.15
= Return on equity (ROE)	**15%**	**-47%**	**18%**	**2%**	**9%**	**155%**	**14%**	**59%**	**18%**	**5%**
Decomposition of RNOA										
Net operating profit margin	7%	-166%	18%	7%	11%	55%	13%	54%	12%	20%
× Net operating asset Turnover	0.65	-2.58	0.55	2.01	0.54	3.20	0.60	1.86	0.65	0.51
= Return on net operating assets	**5%**	**427%****	**10%**	**15%**	**5%**	**175%**	**8%**	**100%**	**9%**	**10%**
Other key profitability ratios										
Gross profit margin	*72%*	*84%*	*70%*	*88%*	*68%*	*85%*	*72%*	*83%*	*73%*	*84%*
EBITDA margin	*3%*	*-195%*	*15%*	*-25%*	*7%*	*81%*	*9%*	*74%*	*9%*	*27%*

*Industry average excluding Moderna **The high RNOA in 2019 is caused by a combination of negative NOA and net losses

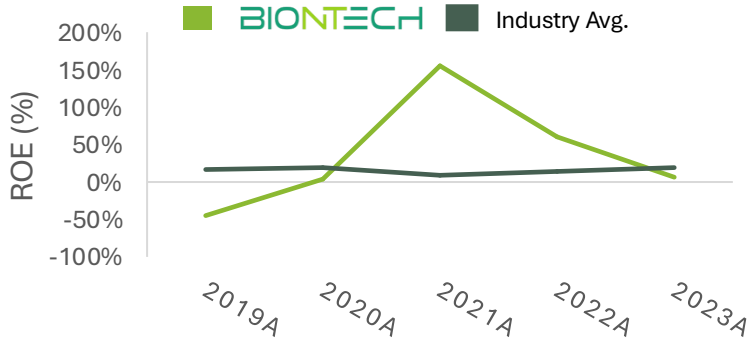
1

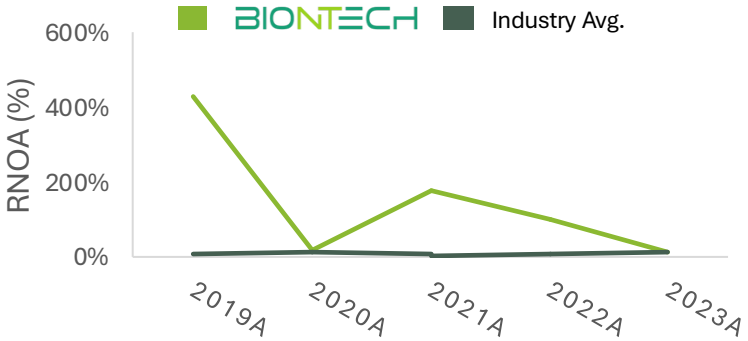
- **ROE moved from 2% in 2020 to 155% in 2021**, mainly due to COVID-19 vaccine revenues
- Higher **net profit margins** and **asset turnover** contributed to this outcome
- As **pandemic demand** declined, ROE normalised to **5% in 2023**
- The **conservative leverage structure** further moderated returns post–vaccine gains

2

- **RNOA** rose from 15% in 2020 to 175% in 2021, primarily due to COVID-19 vaccine revenues
- Higher **operating margins** and **net operating asset turnover** drove this increase
- As **demand normalised,** RNOA settled at **10% in 2023**

3





- BioNTech **outperformed its peers in terms of ROE** in 2021 and 2022 but **returned to below average** in 2023
- **BioNTech's RNOA was above peer average** from 2019 to 2023 due to its high net operating asset turnover

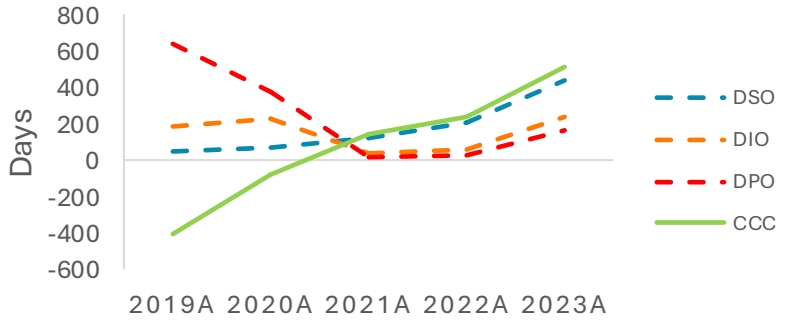
Note: *Industry average excluding Moderna, **The high RNOA in 2019 is caused by a combination of negative NOA and net losses

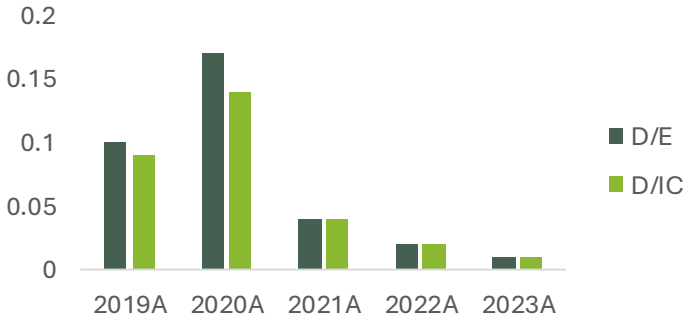
BioNTech ratio analysis: key highlights (2/2)

Post-pandemic investments in operations drive BioNTech's cash conversion cycle

Ratio analysis	Industry avg.* 2019A	BioNTech 2019A	Industry avg.* 2020A	BioNTech 2020A	Industry avg.* 2021A	BioNTech 2021A	Industry avg.* 2022A	BioNTech 2022A	Industry avg.* 2023A	BioNTech 2023A
1 ***Asset management ratios***										
PP&E turnover	2.99	0.82	2.54	2.03	2.71	44.04	3.04	25.41	3.05	4.06
Days' receivables	62.81	51.13	73.84	66.21	70.44	119.01	65.00	203.05	66.18	438.40
Days' inventories	212.65	181.15	254.73	230.15	243.32	35.03	248.46	56.62	250.87	239.27
Days' payables	130.65	643.64	150.96	372.74	139.81	16.22	142.02	21.88	152.56	167.49
Cash conversion cycle	**144.82**	**-411.36**	**177.61**	**-76.38**	**173.95**	**137.83**	**171.45**	**237.79**	**164.48**	**510.18**
2 ***Solvency ratios***										
Debt-to-equity	1.54	0.10	1.23	0.17	0.97	0.04	0.91	0.02	0.96	0.01
Debt-to-invested capital	0.56	0.09	0.52	0.14	0.47	0.04	0.45	0.02	0.45	0.01
Interest coverage (earnings based)	-5.60	87.70	-12.70	0.24	-9.58	-51.32	-11.26	-687.40	-8.37	-51.63
Sustainable growth rate										
ROE	15%	-47%	18%	2%	9%	155%	14%	59%	18%	5%
× Dividend payout ratio	629%	0%	76%	0%	870%	0%	72%	5%	81%	0%
3 ***Sustainable growth rate***	**5%**	**-47%**	**9%**	**2%**	**3%**	**155%**	**8%**	**56%**	**12%**	**5%**

*Industry average excluding Moderna





1
- **PP&E turnover rose from 0.82 in 2019 to 44.04 in 2021**, reflecting COVID-19 output, before falling to **4.06 in 2023**
- **Days' receivables grew** from **51 in** 2019 **to 438 in** 2023, and **days' inventories from 181 to 239**
- As a result, the **cash conversion cycle lengthened from –411 days in 2019 to 510 days in 2023**

2
- **Debt-to-equity** fell from **0.1** in 2019 **to near zero** in 2023, reflecting a **conservative financing approach**
- With **minimal debt, interest coverage remains volatile** due to earnings fluctuations but is not concerning overall
- This **low-leverage strategy reduces financial risk**, though it also limits any **boost to returns from debt financing**

3
- The **sustainable growth rate** rose from **–47%** in 2019 to **155%** in 2021, driven by **higher ROE** and **no dividend payouts** during the COVID-19 vaccine peak
- As **returns stabilised**, the rate **declined** sharply to **5% in 2023**, falling **below** the **12% industry average** as pandemic-related demand waned

Note: *Industry average excluding Moderna

Accounting adjustments

Overview over potential accounting adjustments

We did not make any accounting adjustments to BioNTech's financial statements

Adjustments to BioNTech's financial statements (1/2)

Revenue, allowances and impairment adjustments

- There are no signs that adjustments should be made for too aggressive (or conservative) recognition of revenues, allowances or impairments.

Post-employment benefit adjustment

- We did not make any adjustments as the liabilities arising from the very few pension agreements are unsubstantial and not value-relevant.

R&D adjustment

- Due to a lack of reported data regarding in-process R&D and capitalisation rates, we did not make an R&D adjustment.

Adjustments to BioNTech's financial statements (2/2)

Depreciation adjustment

- The composition and implied useful life of assets greatly differ between BioNTech and its peers.
- The implied useful life of non-current operating assets is roughly in line with that of Moderna, the peer most closely related to BioNTech. Thus, we did not make any adjustments to BioNTech's depreciation charges.

Adjustments for non-recurring items

- The fact that BioNTech's past financials only have limited predictive and informative value for the future due to the COVID-19 peak is considered in the forecast but not reflected in the form of any accounting adjustments.

	BIONTECH		AstraZeneca		moderna		sanofi		CSK		Takeda	
Depreciation policy adjustment	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**	**2020A**	**2023A**
PP&E as a % of PP&E + intangibles	62%	79%	22%	15%	96%	69%	16%	15%	23%	34%	16%	19%
Depreciation (% of non-current operating assets)	16%	18%	6%	5%	14%	26%	4%	4%	5%	4%	2%	1%
Implied useful life	6.14	5.71	15.74	18.92	7.13	3.80	22.95	27.51	18.93	25.53	55.05	82.30

Valuation

Selected public comparable companies

BioNTech trades above its peer group for forward-looking multiples

BIONTECH

Company name	Price close	Shares oust. (m)	Market Cap	Enterprise Value	Revenue (LTM)	EV / Revenue (LTM)	Revenue (NTM)	EV / Revenue (NTM)
BioNTech SE	**93.14**	**240**	**22,351**	**6,513**	**2,751**	**2.37x**	**2,118**	**3.08x**
Selected Biotech peers								
moderna	31.62	387	12,227	4,108	2,992	1.37x	2,077	1.98x
AstraZeneca	136.83	1,551	212,172	236,051	49,995	4.72x	54,520	4.33x
CSK	17.74	4,077	72,335	86,762	37,066	2.34x	39,343	2.21x
sanofi	103.88	1,224	127,153	137,860	44,286	3.11x	46,188	2.98x
Takeda	27.93	1,529	42,717	73,061	27,970	2.61x	28,216	2.59x
Min			12,227	4,108	2,751	1.37x	2,077	1.98x
Median			57,526	79,911	32,518	2.49x	33,779	2.79x
Average			81,492	90,726	27,510	2.75x	28,744	2.86x
Max			212,172	236,051	49,995	4.72x	54,520	4.33x

Relative Value	
EV / Revenue (NTM)	2.79
Revenue (NTM)	2,800
Enterprise Value	**7,804**
Net Debt	(15,838)
Equity Value	**23,642**
Implied price p. share	**98.52**

Source(s): Capital IQ as of 25th of March 2025

BioNTech WACC analysis

Cost of capital is driven by the cost of equity due to low leverage

Input	Value	Comment
Risk-free rate	4.35%	Yield on 10y U.S. government bond
Equity beta	1.01	Based on 5y-monthly regression (against S&P 500)
Equity risk premium (ERP)	4.33%	Implied current U.S. ERP
Cost of equity	**8.70%**	
Pre-tax cost of debt	7.43%	Based on AlphaSpread security analysis
Marginal tax rate	30.00%	German corporate tax rate
Post-tax cost of debt	**5.20%**	
Debt / value	0.97%	Current capital structure
Equity / value	99.03%	
WACC	**8.67%**	

Sensitising WACC based on capital structure and levered beta

		Debt / value				
		1%	**5%**	**9%**	**13%**	**17%**
Equity beta	**0.81**	7.81%	7.70%	7.60%	7.49%	7.39%
	0.91	8.24%	8.12%	7.99%	7.87%	7.75%
	1.01	8.67%	8.53%	8.39%	8.25%	8.11%
	1.11	9.10%	8.94%	8.78%	8.62%	8.47%
	1.31	9.95%	9.76%	9.57%	9.38%	9.19%

Sensitising WACC based on capital structure and cost of debt

		Debt / value				
		1%	**5%**	**9%**	**13%**	**17%**
Pre-tax cost of debt	**0.03**	8.64%	8.39%	8.14%	7.88%	7.63%
	0.05	8.65%	8.46%	8.26%	8.07%	7.87%
	0.07	8.67%	8.53%	8.39%	8.25%	8.11%
	0.09	8.68%	8.60%	8.51%	8.43%	8.34%
	0.11	8.69%	8.67%	8.64%	8.61%	8.58%

Source(s): Capital IQ as of 25th of March 2025, AlphaSpread as of 25th of March.

PV of FCF analysis

Negative FCF driven by high R&D investments leaving all of terminal value in exit multiple

BIONTECH

PV of FCF Valuation	2019A	2020A	2021A	2022A	2023A	2024E	2025E	2026E	2027E	2028E
Revenue	**109**	**482**	**18,977**	**17,311**	**3,819**	**2,800**	**2,250**	**2,178**	**1,992**	**2,056**
Cost of goods sold (COGS)	(17)	(59)	(2,912)	(2,995)	(600)	(429)	(345)	(334)	(305)	(315)
Gross profit	**91**	**423**	**16,065**	**14,316**	**3,219**	**2,371**	**1,905**	**1,844**	**1,686**	**1,741**
Gross margin (%)	*84%*	*88%*	*85%*	*83%*	*84%*	*85%*	*85%*	*85%*	*85%*	*85%*
Operating income/expenses	**(273)**	**(505)**	**(781)**	**(1,673)**	**(2,529)**	**(2,593)**	**(2,931)**	**(2,773)**	**(2,646)**	**(2,679)**
SG&A	(48)	(109)	(327)	(541)	(558)	(409)	(329)	(318)	(291)	(300)
R&D	(227)	(645)	(949)	(1,537)	(1,783)	(2,240)	(2,647)	(2,499)	(2,395)	(2,420)
Other operating income/expenses	2	248	495	405	(188)	56	45	44	40	41
Operating profit	**(182)**	**(82)**	**15,284**	**12,643**	**690**	**(222)**	**(1,026)**	**(930)**	**(959)**	**(939)**
Operating margin (%)	*-167%*	*-17%*	*81%*	*73%*	*18%*	*-8%*	*-46%*	*-43%*	*-48%*	*-46%*
Taxes	0	161	(4,754)	(3,520)	(256)	67	308	279	288	282
NOPAT	**(183)**	**78**	**10,531**	**9,124**	**435**	**(156)**	**(718)**	**(651)**	**(672)**	**(657)**
Plus: D&A						134	108	105	96	99
Less: Capex						(168)	(135)	(131)	(120)	(123)
Less: Changes in NWC						1,828	95	12	32	(11)
Free cash flow						**1,638**	**(651)**	**(665)**	**(664)**	**(693)**
Discounted free cash flow						**1,508**	**(551)**	**(518)**	**(476)**	**(457)**
Sum	*(495)*									

Negative FCF due to high expected R&D spending

Sensitising WACC against terminal value multiple

		WACC				
		6.7%	**7.7%**	**8.7%**	**9.7%**	**10.7%**
EV / NTM revenue	**0.8x**	50.32	50.31	50.30	50.29	50.28
	1.8x	56.84	56.53	56.24	55.96	55.70
	2.8x	63.35	62.75	62.17	61.63	61.12
	3.8x	69.87	68.96	68.11	67.30	66.54
	4.8x	76.38	75.18	74.05	72.97	71.96

Assumptions	
Tax Rate	30.00%
WACC	8.67%
Perpetural Growth Rate	2.00%
EV / NTM Revenue Multiple	2.79x
Current Price	93.14
Shares Outstanding (m)	239.97
Cash	11,663.70
Debt	219.10
Terminal Revenue Growth Rate	5.00%

Terminal Value	
Revenue NTM	2,159
EV / NTM Revenue Multiple	2.79x
Terminal Value	6,016
Present Value of Terminal Value	3,970

Intrinsic Value	
Enterprise Value	3,475
Plus: Cash	11,664
Less: Debt	219
Equity Value	**14,920**
Implied Price p. share	**62.17**
Difference to actual	*-33.2%*

Source(s): Capital IQ as of 25th of March 2025

PV of ANOPAT analysis

ANOPAT leads to higher valuation due to capital charge with favourable low WACC

BIONTECH

PV of ANOPAT Valuation	2019A	2020A	2021A	2022A	2023A	2024E	2025E	2026E	2027E	2028E
NOPAT						(156)	(718)	(651)	(672)	(657)
Beginning net operating assets (t-1)						4,292	2,498	2,431	2,444	2,436
Capital charge						372	217	211	212	211
ANOPAT						(528)	(935)	(862)	(884)	(869)
PV of ANOPAT						**(486)**	**(792)**	**(672)**	**(634)**	**(573)**
Sum	*(3,156)*									
						(1,993)	(241)	(154)	(158)	(116)

Terminal value	
Revenue NTM	2,159
EV / NTM revenue multiple	2.79
Terminal value	6,016
Present value of terminal value	3,970
Intrinsic value	
Enterprise value	5,107
Plus: Cash	11,664
Less: Debt	219
Equity value	**16,551**
Implied price per share	**68.97**
Difference to actual	*-25.9%*

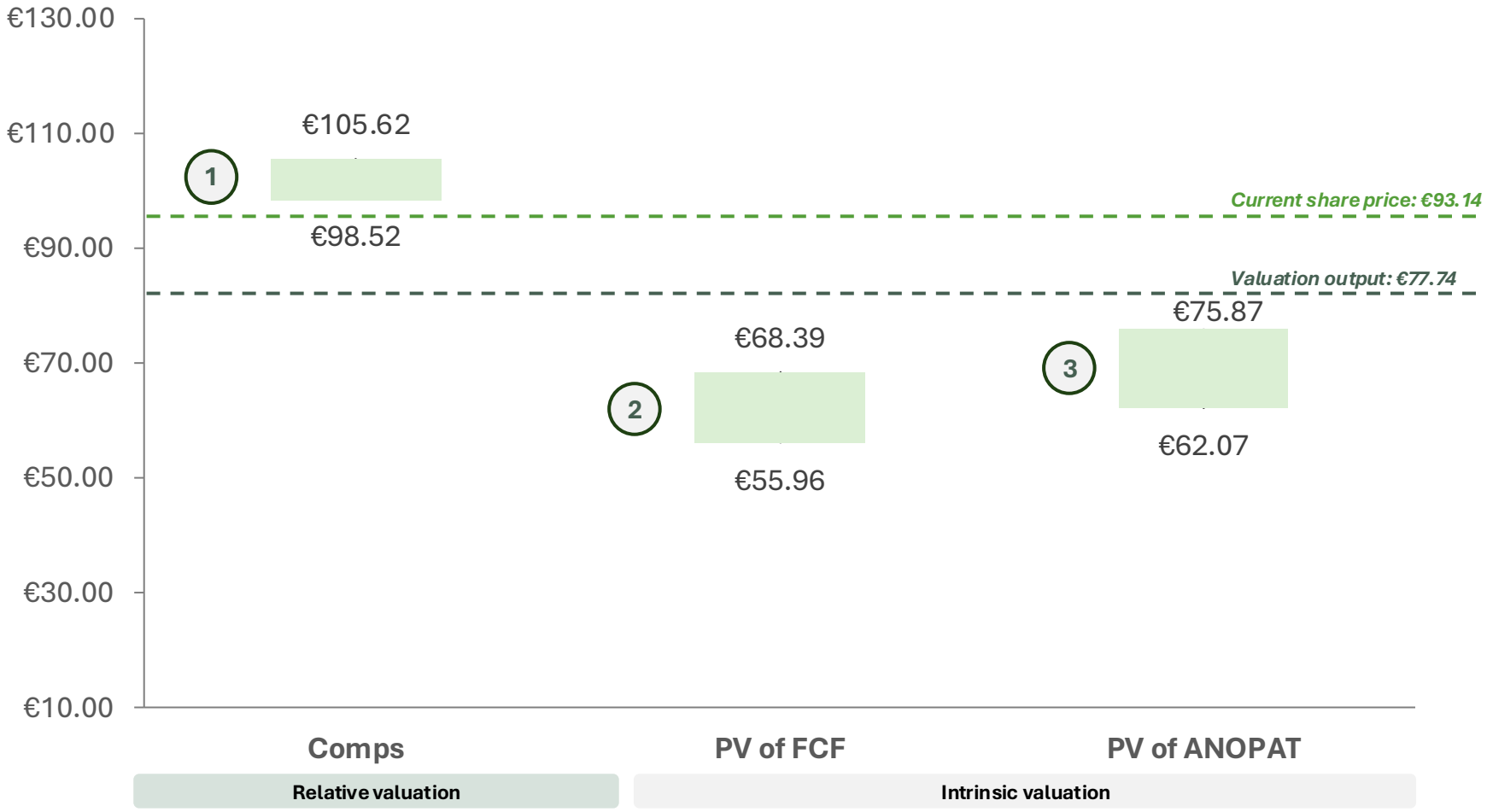
Sensitising WACC against terminal value multiple

EV / NTM Revenue	WACC 6.7%	7.7%	8.7%	9.7%	10.7%
0.8x	57.78	57.43	57.10	56.78	56.49
1.8x	64.30	63.65	63.04	62.46	61.91
2.8x	70.81	69.87	68.97	68.13	67.33
3.8x	77.33	76.08	74.91	73.80	72.75
4.8x	83.84	82.30	80.85	79.47	78.17

Source(s): Capital IQ as of 25th of March 2025

Valuation overview

Valuation share price below actual due to strong emphasis on intrinsic valuation



1

- Increased comparable company valuation due to **high EV / NTM Revenue multiple outcomes** between **2.5x – 2.8x**
- Revenue multiple as it captures **growth potential despite volatile earnings**

2

- Valuation between €55.9 and €68.4 due to **negative FCF during forecast period**
- **High R&D spending for upcoming product pipeline development** as main contributor to negative FCF

3

- Slightly **higher ANOPAT valuation compared to FCF**, but still below comparable valuation and current share price
- Main **value driver lays in terminal value**, due to upcoming pipeline developments

Source(s): Capital IQ as of 25th of March 2025

Valuation conclusions

Internal analysis leads to a conservative sell recommendation

BIONTECH

#1	**Company performance outlook uncertain**	BioNTech's future performance is highly dependent on regulatory approval of the BNT122 & BNT116 vaccines which would create a competitive edge (as for COVID-19)
#2	**Discounted cash flow analysis**	Implied equity valuation leads to ~30% discount when applying conservative exit multiple (NTM EV/Revenue) and current capital structure going forward
#3	**Public comps**	BioNTech currently trades above industry avg. implying strong investor sentiment towards product differentiation and probability of success of future vaccines
#4	**Industry outlook**	Biopharmaceuticals are shifting towards precision medicine, mRNA technologies, and immunotherapies, as long-term growth markets but demand high R&D investment

Company analysis and valuation

BIONTECH

AC444 – Summative Assessment by Group 1, Seminar 1